

Delivery Customer Support Chatbot Using BERT, Text-CNN, and Reinforcement Learning

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ABSTRACT

This study presents a hybrid customer support assistant for handling delivery-related queries using natural language processing and reinforcement learning. The system utilizes a fine-tuned BERT model for intent classification, with a Text-CNN implemented as a baseline for comparison. A Multi-Armed Bandit with an epsilon-greedy strategy is integrated to improve response selection through adaptive learning. The models are trained on a publicly available e-commerce customer support dataset and evaluated using accuracy and F1-score. Results show that BERT outperforms Text-CNN in classification performance, while the reinforcement learning component enhances response selection behavior over time. Additional analyses indicate that performance is influenced by dataset coverage and sensitivity to vocabulary variations. The system demonstrates the effectiveness of combining deep learning and lightweight reinforcement learning for domain-specific customer support, while highlighting limitations in handling ambiguous queries and multi-turn interactions.

CCS Concepts

• **Computing methodologies**→ **Natural language processing** • **Computing methodologies**→ **Supervised learning by classification** • **Computing methodologies**→ **Reinforcement learning** • **Information systems**→ **Question answering**

Keywords

natural language processing; intent classification; BERT; Text-CNN; reinforcement learning; multi-armed bandit; customer support chatbot; machine learning

1. PROBLEM AND MOTIVATION

1.1 Problem Statement

This project aims to develop an intelligent customer support chatbot for delivery service operations. The system focuses on handling delivery-related inquiries such as package tracking, delivery issues, missing items, and shipping costs. Automating these common support tasks can improve response time, reduce workload on human agents, and enhance customer experience.

1.2 Success Metrics

The performance of the system is evaluated primarily using intent classification metrics, including accuracy, precision, recall, and F1 score. The F1 score is used as the main metric as it balances precision and recall across all intent classes. In addition, the effectiveness of the reinforcement learning component is assessed through simulated reward signals representing successful response selection.

1.3 Constraints

The system is limited to delivery-related customer support queries and does not handle other types of inquiries such as billing or product returns. The dataset consists of structured and synthetic customer queries, which may not fully reflect real-world language variations. Furthermore, the reinforcement learning component relies on simulated rewards rather than real user feedback, which may limit its real-world applicability.

2. DATASET

The dataset used in this study was obtained from the Hugging Face dataset repository, specifically the Bitext Retail E-commerce LLM Chatbot Training Dataset. This dataset is publicly available under the Community Data License Agreement – Sharing, Version 1.0, and was selected due to its relevance to customer support interactions in the e-commerce domain. It contains annotated customer queries paired with corresponding intent labels, making it suitable for supervised learning in intent classification tasks.

Since the dataset includes a wide range of e-commerce intents, a filtering process was applied to retain only those relevant to the delivery service scope of the system. The selected intents include `track_order`, `track_delivery`, `delivery_issue`, `delivery_time`, `damaged_delivery`, `missing_item`, and `shipping_costs`. Intents outside this scope were removed to ensure that the model is trained on focused and domain-specific data. In addition, duplicate entries were eliminated to improve data quality and reduce redundancy in the dataset.

Prior to model training, the dataset underwent preprocessing to standardize and prepare the data. Text inputs were cleaned and formatted to ensure consistency across all samples. Tokenization

was applied to convert text into numerical representations, using the BERT tokenizer for both the transformer-based BERT model and the Text-CNN model to maintain consistency in input representation. Furthermore, intent labels were encoded into numerical form to enable supervised learning. These preprocessing steps ensure that the dataset is compatible with the deep learning models used in this study.

The processed dataset was then used for training the intent classification models. Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score, with the F1 score serving as the primary evaluation metric. This approach provides a reliable measure of the models' effectiveness in correctly classifying delivery-related customer queries.

3. TRAINING DETAILS

The models are trained on a preprocessed intent classification dataset, where labels are encoded into numeric IDs to enable supervised learning. A fixed random seed (42) is used to ensure reproducibility.

For the BERT model, fine-tuning is conducted using the Hugging Face Trainer API. The model uses the bert-base-uncased tokenizer with a maximum sequence length of 32, applying padding and truncation. Training is performed with a learning rate of 2×10^{-5} , batch size of 32, and 3 epochs. Mixed precision training (fp16) is enabled for faster computation on GPU. Evaluation and checkpoint saving are performed at each epoch, with the best model automatically selected at the end. Performance is measured using accuracy, precision, recall, and F1-score.

For the Text-CNN model, training is implemented in PyTorch using an embedding dimension of 128 and three convolutional layers with kernel sizes 3, 4, and 5 (100 filters each). The model is trained with a batch size of 32, learning rate of 1×10^{-3} , and 3 epochs using the Adam optimizer and cross-entropy loss. Input text is tokenized using the same BERT tokenizer with a maximum length of 50. Model performance is evaluated using accuracy and F1-score.

The RL component is implemented as a Multi-Armed Bandit using an epsilon-greedy strategy. An exploration rate of 0.2 is used, where the agent selects a random action 20% of the time and exploits learned values otherwise. Action-value estimates are updated incrementally based on observed rewards, allowing the agent to improve response selection over time without requiring full retraining.

Training is performed on a GPU-enabled environment when available, with CPU fallback. Reproducibility is ensured through fixed random seeds, consistent preprocessing, saved model checkpoints, and preserved tokenizer configurations, enabling reliable comparison and reuse of results.

4. MODELING

The system adopts a hybrid architecture that combines Natural Language Processing (NLP) and Reinforcement Learning (RL) to effectively handle delivery-related customer support queries. For intent classification, a fine-tuned BERT model is used as the primary architecture due to its strong contextual understanding of language. Its bidirectional nature allows it to interpret user queries more accurately, especially when messages are short or ambiguous.

To complement this and satisfy the CNN requirement, a Text-CNN model is implemented as a comparative approach. The

Text-CNN captures local textual patterns using convolutional filters and offers faster computation with competitive performance for sentence classification tasks. This enables a meaningful comparison between transformer-based and convolution-based models.

An RL agent is incorporated to improve response selection. It receives the predicted intent and determines the most appropriate response based on a learned policy, using a reward mechanism that considers response relevance and successful interaction outcomes.

The overall workflow follows a pipeline: user input is first processed by the NLP model (BERT or Text-CNN) for intent prediction, then passed to the RL agent for decision-making, and finally mapped to the appropriate response. This modular design ensures both accurate understanding and adaptive response generation.

5. EVALUATION

5.1 Evaluation Metrics

The performance of the intent classification models was evaluated using accuracy, precision, recall, and F1 score. Accuracy measures the proportion of correctly classified samples, while precision and recall evaluate the model's performance in identifying each intent class. The F1 score, which is the mean of precision and recall, was used as the primary evaluation metric as it provides a balanced measure of model performance across all classes. All metrics were computed using a weighted average to account for the distribution of samples across different intent categories.

5.2 Baseline Models

The Text-CNN model was used as a baseline deep learning approach for intent classification. This model applies convolutional filters to text embeddings to capture local patterns in customer queries. The BERT model was evaluated as a more advanced transformer-based approach, allowing for comparison between convolutional and contextual language models.

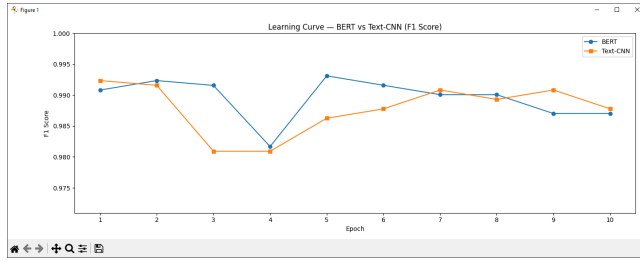
5.3 Model Comparison

The Text-CNN and BERT models were evaluated under the same training conditions using both 1 and 3 training epochs.

Model	Epoch	Accuracy	F1 Score
TEXT-CNN	1	99.08%	99.08%
BERT	1	99.13%	99.08%
TEXT-CNN	3	99.16%	99.16%
BERT	3	99.18%	99.16%

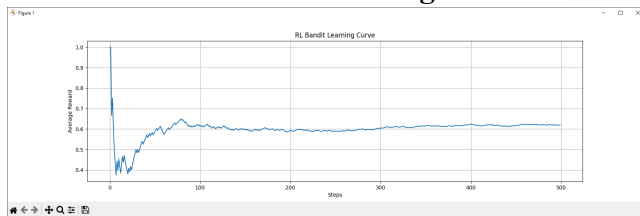
The results show that both models achieve very similar performance, with accuracy and F1 scores exceeding 99% in all configurations. While BERT slightly outperforms the Text-CNN, the difference is minimal. This indicates that for short, structured customer queries, simpler convolutional models are capable of achieving performance comparable to transformer-based models. Given the small performance gap, the Text-CNN offers a more computationally efficient alternative, while BERT provides marginally better accuracy.

5.4 Model Learning Curves

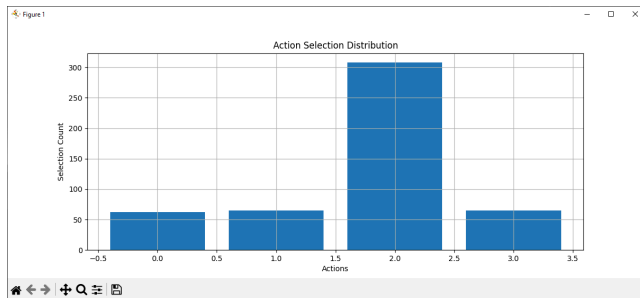


Both models maintain F1 scores above 0.98 throughout all epochs, confirming strong and stable performance on the intent classification task. The dips observed around epochs 3–5 in both models may reflect the optimization process navigating a local minimum before recovering. Neither model shows signs of significant overfitting, as performance does not degrade consistently toward the later epochs. Overall, BERT demonstrates slightly more stable learning while Text-CNN shows greater early volatility but comparable final performance.

5.5 Reinforcement Learning Results



This graph shows the learning behavior of the reinforcement learning agent. Initially, the average reward fluctuates due to high exploration ($\epsilon = 0.5$). As the number of steps increases, the agent gradually learns to favor more rewarding actions, resulting in a more stable and higher average reward. This demonstrates the effectiveness of the epsilon-greedy strategy in improving decision-making over time. Although convergence is slower due to the relatively high exploration rate, it ensures broader exploration of possible actions.



This illustrates the distribution of action selections by the reinforcement learning agent. It can be observed that one action is selected significantly more frequently than others, indicating that the agent has successfully identified and exploited the most rewarding option. The presence of selections for other actions reflects the exploration component of the epsilon-greedy strategy ($\epsilon = 0.5$), where the agent continues to explore alternative actions. This demonstrates a balance between exploration and exploitation during learning.

However, this evaluation is based on simulated rewards and does not fully reflect real user interactions. Additionally, the

multi-armed bandit approach does not consider sequential context, limiting its ability to model multi-turn conversations.

5.6 Ablation Study

5.6.1 Model Architecture

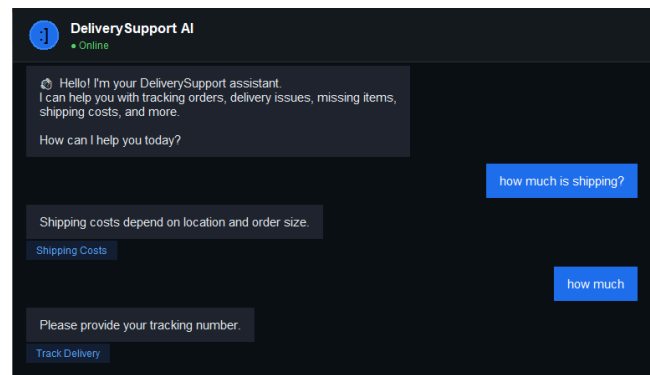
The first ablation compares two deep learning architectures for intent classification: a Text-CNN model and a BERT-based model. Both models were trained on the same dataset using identical preprocessing steps and evaluation metrics to ensure a fair comparison. The results show that both models achieved high performance, with BERT slightly outperforming the Text-CNN. However, the difference in accuracy and F1 score was minimal, indicating that convolutional models can perform comparably to transformer-based models for short and structured customer queries.

5.6.2 Number of Training Epochs

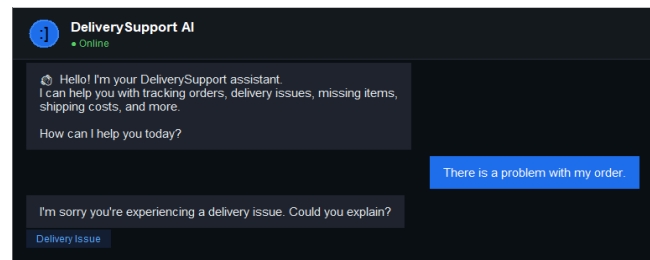
The second ablation examines the effect of training duration by comparing model performance after 1 epoch and 3 epochs. Both the Text-CNN and BERT models were evaluated under these settings. The results show a slight improvement in performance when increasing the number of training epochs. However, both models already achieved high accuracy after a single epoch, suggesting that the dataset is relatively easy to learn and that additional training provides diminishing returns.

5.7 Error Analysis

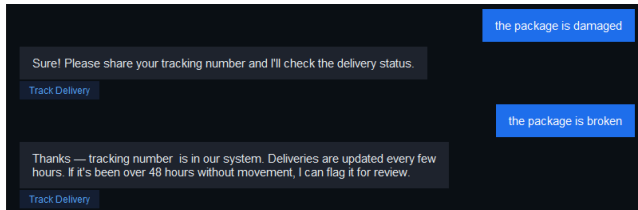
An error analysis was conducted to examine cases where the model produced incorrect predictions. Although the overall performance of both models was high, a small number of misclassifications were observed.



Some errors were observed when user queries lacked sufficient context. For example, the query “how much” was incorrectly classified as *track_delivery*, while the more specific query “how much is shipping” was correctly identified as *shipping_costs*. This suggests that the model relies heavily on explicit keywords and struggles with ambiguous or incomplete inputs.



Additionally, some misclassifications were caused by ambiguous or vague user queries that lacked specific keywords. For example, general statements such as “there is a problem with my order” can correspond to multiple intent categories, making accurate classification more difficult.



Model performance is sensitive to vocabulary coverage; synonym variations not present in the training data (e.g., ‘damaged’ vs. ‘broken’) may lead to misclassification, highlighting the importance of dataset diversity.

These findings suggest that while the model performs well on clear and structured queries, it may face challenges when handling ambiguous or overlapping intent categories.

6. ETHICS & POLICY

The system is designed for automated delivery-related customer support, but several risks must be considered. One key risk is incorrect intent classification, which may lead to irrelevant or misleading responses. To mitigate this, the model is evaluated using accuracy and F1-score, and the RL component is designed to improve response selection over time. Additionally, the system can be limited to predefined response templates to reduce the chance of harmful or unsupported outputs.

In terms of privacy, the system processes user-generated text, which may contain sensitive information. To address this, only publicly available or properly licensed datasets are used during training, and no personally identifiable information (PII) is stored or retained in the system. Any deployed version should include input filtering or anonymization to further protect user data.

For fairness, biases may arise from the training dataset, especially if certain query types or language styles are overrepresented. To mitigate this, the dataset is reviewed for balance across intent categories, and performance metrics are monitored per class to detect uneven behavior. However, the system may still perform less accurately on uncommon or ambiguous queries.

The intended use of the system is to assist with common delivery-related inquiries such as tracking orders or reporting issues. It is not intended to replace human support in complex, sensitive, or high-stakes situations.

Key limitations include dependence on training data quality, restricted understanding outside predefined intents, and the simplicity of the RL approach (multi-armed bandit), which does not model long-term conversational context. Users should be informed that responses are AI-generated and may not always be fully accurate, and a fallback to human support should be available when needed.

7. MODEL CARD

The model is a hybrid customer support assistant combining BERT-based intent classification with a Multi-Armed Bandit for adaptive response selection. BERT (bert-base-uncased) serves as the primary classifier, fine-tuned on delivery-related customer queries across seven intent categories: order tracking, delivery issues, missing items, shipping costs, and damaged delivery. A

Text-CNN model is included as a lightweight baseline for comparison. Both models were trained with a maximum sequence length of 32 tokens, a batch size of 32, and evaluated over 1 and 3 epochs.

The system achieves a best accuracy of 99.18% and an F1 score of 99.16% using BERT at 3 epochs. The Multi-Armed Bandit component uses an epsilon-greedy strategy ($\epsilon = 0.5$) for response selection, learning to favor higher-reward responses over time through incremental action-value estimation. Training data was sourced from the Bitext retail and e-commerce dataset on Hugging Face, licensed under the Community Data License Agreement – Sharing, Version 1.0.

The system is intended for academic use in automating delivery-related support queries. It is not designed for open-domain dialogue, multi-turn conversations, or handling sensitive personal data. Key limitations include reliance on simulated rewards for RL evaluation, sensitivity to vocabulary coverage, and classification restricted to predefined intent categories.

8. HOW TO RUN

The system is designed to be fully reproducible using a single command after completing the required environment setup.

8.1 Environment Setup

The project requires Python 3.10 or higher. Dependencies are installed using the provided requirements file, while PyTorch is installed separately depending on hardware capability (GPU or CPU). The installation can be completed using the following commands:

```
python -m pip install torch==2.5.1+cu121 --index-url
https://download.pytorch.org/whl/cu121
python -m pip install -r requirements.txt
```

For systems without GPU support, the CPU version of PyTorch can be installed instead.

8.2 Dataset Preparation

The system expects the dataset to be located at data/processed/intent_dataset.csv. If the dataset is not available, it can be generated by running: commands:

```
python src/data_pipeline.py
```

8.3 One-Command Reproduction

After installation and dataset preparation, the entire pipeline can be executed using a single command:

```
run_all.bat
```

This command performs model training (BERT and Text-CNN), runs the reinforcement learning evaluation, and generates the corresponding performance graphs.

8.4 Demo Instructions

For demonstration purposes, users may execute the system using the provided commands. To train the models and generate evaluation outputs (e.g., learning curves and action distribution graphs), run:

```
run_all.bat
```

This will produce evaluation results saved in the experiments/ directory, which can be used to verify model performance.

To interact with the system as a chatbot, run the graphical user interface:

```
python chatbot_ui.py
```

This launches the chatbot application, allowing users to input queries and receive responses based on the trained models.

9. REFERENCES

- [1] Bitext. 2024. Bitext Retail E-commerce LLM Chatbot Training Dataset. Hugging Face.
<https://huggingface.co/datasets/bitext/Bitext-retail-ecommerce-llm-chatbot-training-dataset>